



Exploring IPv6

The end of the Internet as we
know today?



Gratien D'haese
IT3 Consultants
gratien.dhaese@it3.be



Conclusion

- The end of the Internet as we know today?
 - IPv4 address space is getting scarce
 - IPv4 will still be available for a long time
 - IPv6 is getting slowly deployed
 - IPv6 will boost from this year on
 - Not because we like it, but because we have no choice
 - No need to be afraid of IPv6 (after this talk :)
 - Dual stack with IPv4, or 6to4 tunnels



Abbreviations

- IPv4/6: Internet Protocol 4/6
- ISC: Internet Systems Consortium
- IANA: Internet Assigned Numbers Authority
- RIR: Regional Internet Number Registries
- CIDR: Classless Inter-domain Routing
- NAT: Network Address Translation
- AS: Autonomous System



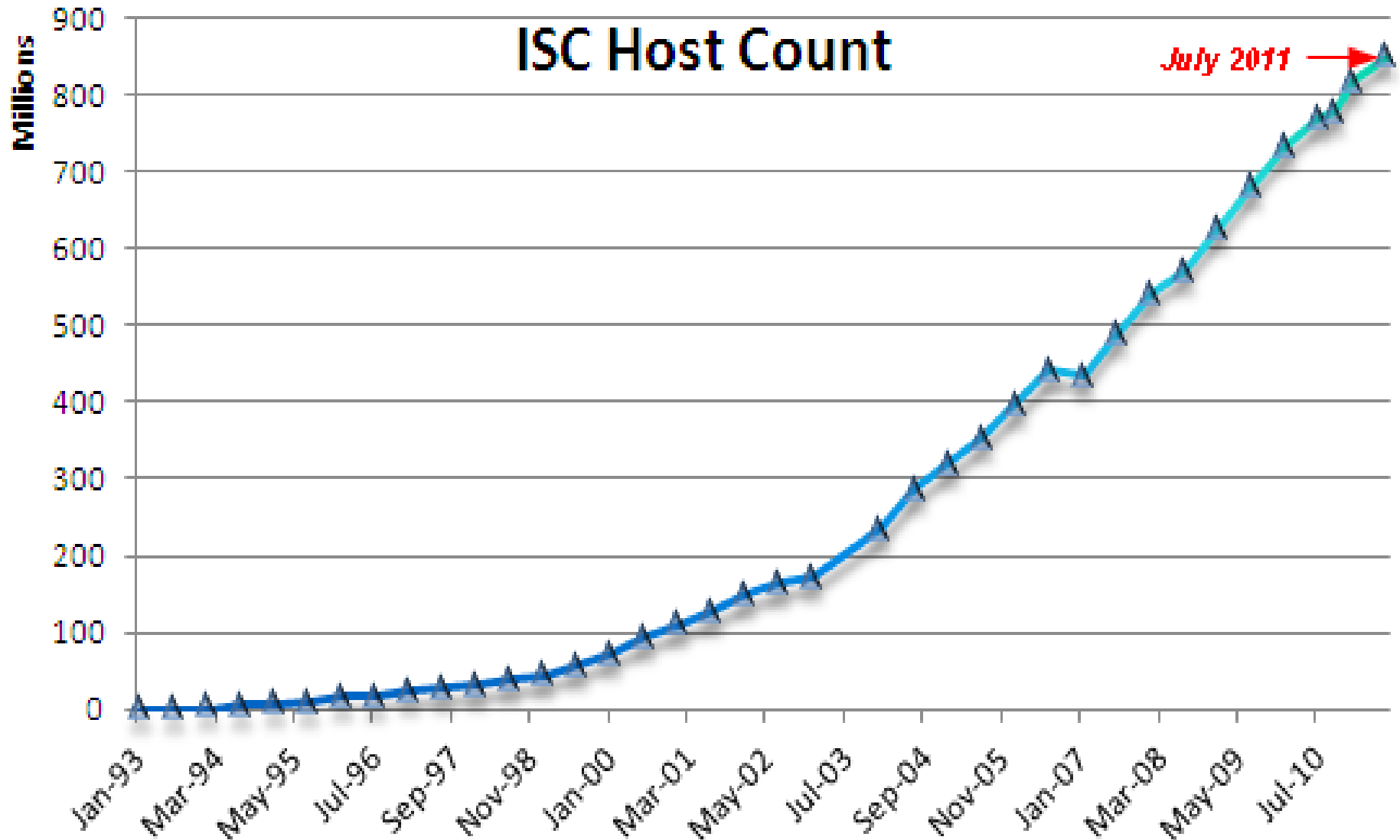
IPv6 history

- Designed in 1994 [RFC 1752 and many more]
- In the nineties estimated run-out of IPv4 addresses was expecting between 2000-2008
- The usage of CIDR and NAT slowed down the depletion of IPv4 addresses, but also
 - The dot com crisis, and
 - Financial crisis in 2008-2009
- The Internet still grows rapidly (mobile devices,...)



The IPv4 host count 'till today

(data coming from ISC)





IPv4 Address Space

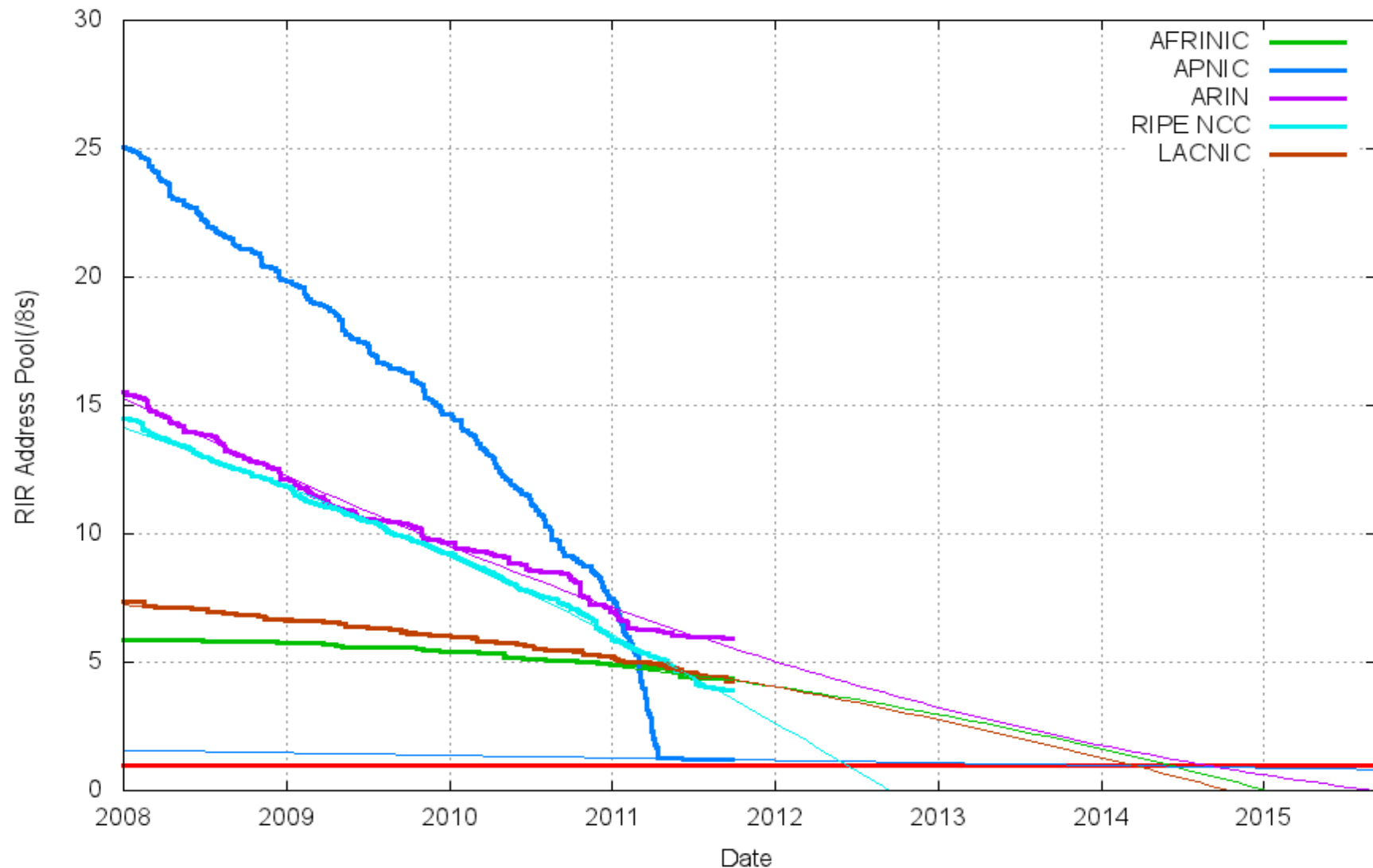
- 32-bit number $\Rightarrow 2^{32}$ (4.294.967.296)
- 4 dotted decimal notation, e.g. 18.2.45.78
- Divided into classes
 - A Class: 8-bit network (128 * 16,8 million)
 - B Class: 16-bit network (16.384 * 65.536)
 - C Class: 24-bit network (2 million * 256)
- 70% of A and B Classes are allocated to big companies and incredible under-used (approx. 3 billion addresses wasted)



IPv4 Depletion rate

www.potaroo.net/tools/ipv4/

RIR IPv4 Address Run-Down Model





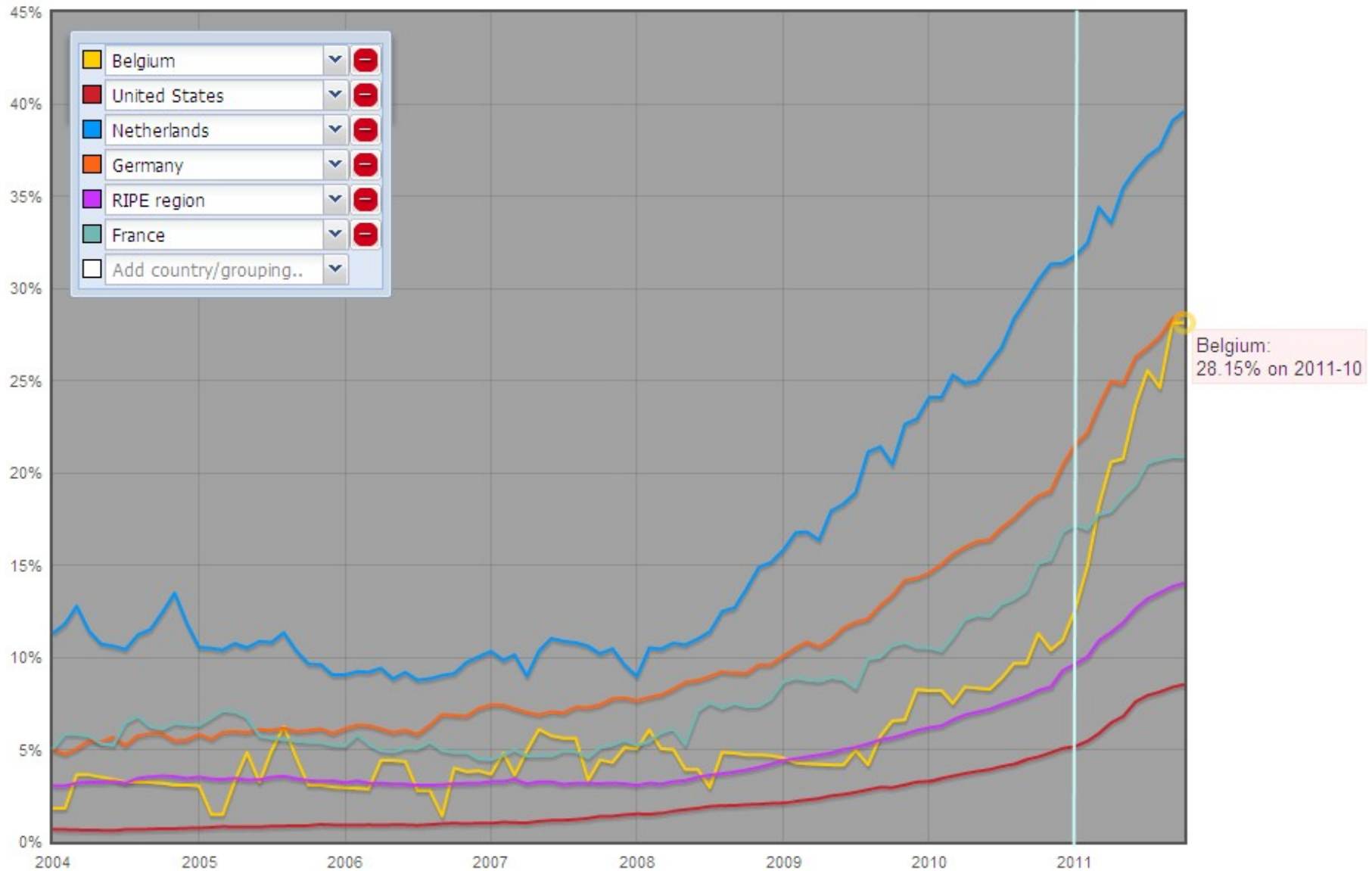
IPv6 history

- Backbone routers (vendors): took time to become IPv6 ready
 - Today these limitations are behind us
 - But, are all ISP's capable for serving IPv6 traffic?
- The main Operating Systems (Linux, Mac OS/X and Windows) now support IPv6
- IPv6 has been implemented more widely in Europe and Asia than in the USA.



IPv6 enabled ASs in global routing

<http://v6asns.ripe.net/>





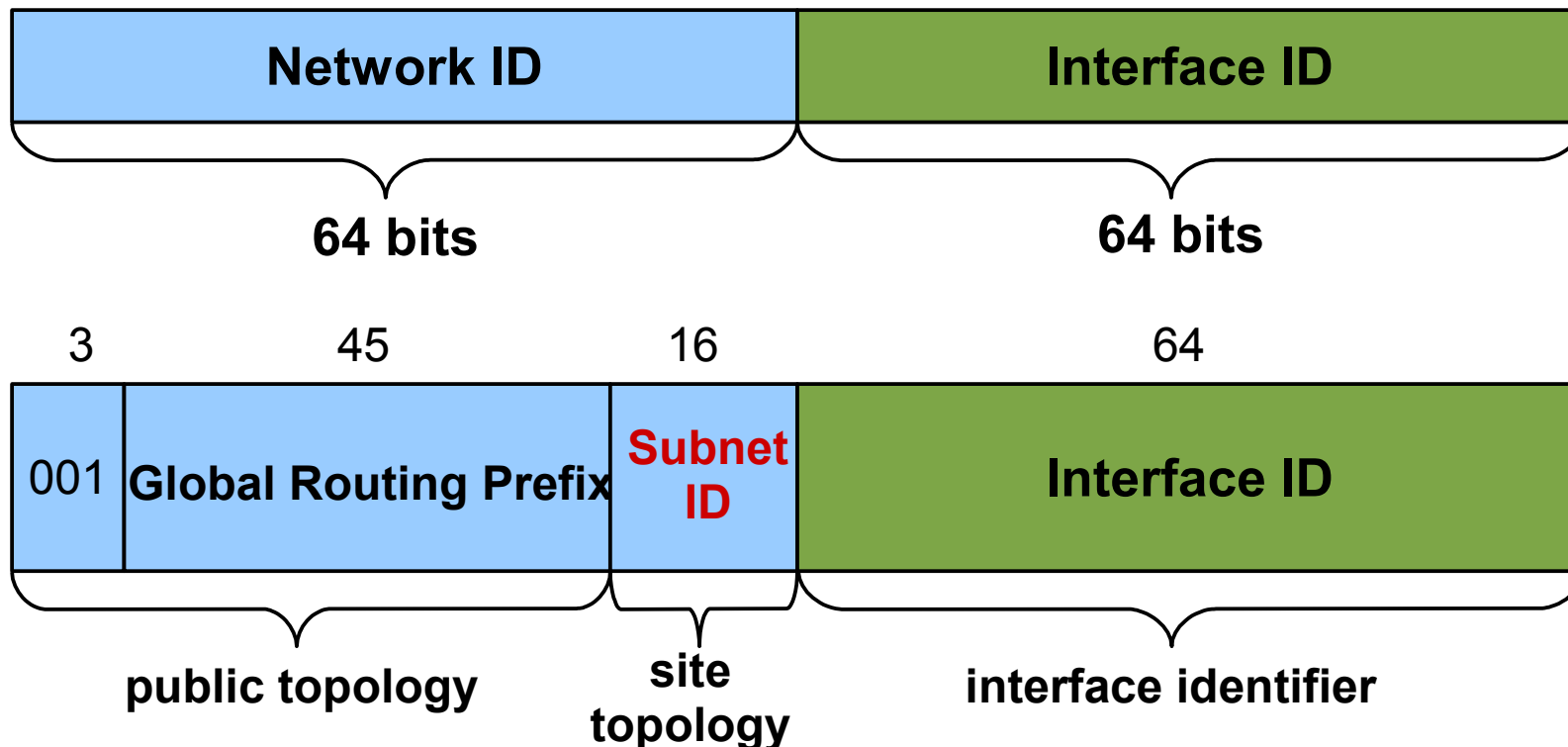
Is your ISP IPv6 ready ?

- Have a look at
 - <http://ripeness.ripe.net/4star/BE.html>
 - <http://www.vyncke.org/ipv6status/detailed.php?country=be&type=ISP>
- Most ISPs will deliver IPv6 to home consumers not before 2012 (or 2013?) ...
- Around 48% ISPs can provide IPv6 addresses
 - See <http://ripeness.ripe.net/pies.html>
 - Mostly through IPv6-to-IPv4 tunneling
 - One year ago it was only 31%



IPv6 Addressing

- $2^{128} = 3.4 \times 10^{38}$ addresses (128 bits!!)
= 340.282.366.920.938.463.463.374.607.431.768.211.456
- IPv6 address is divided into





IPv6 Addressing (cont.)

- Notation
 - IPv6 address written as eight groups of four hexadecimal digits
 - 2001:0db9:85a6:07c4:1243:8a81:0301:7351
 - Leading zeros may be dropped
 - 2001:9a03:0000:12c2:0000:0000:0fa1:0001
 - 2001:9a03:0:12c2:0:0:fa1:1
 - Up to one double colon substitution is permitted
 - 2001:9a03:0:12c2::fa1:1
 - :: means one or more groups of 16 bits of zeroes

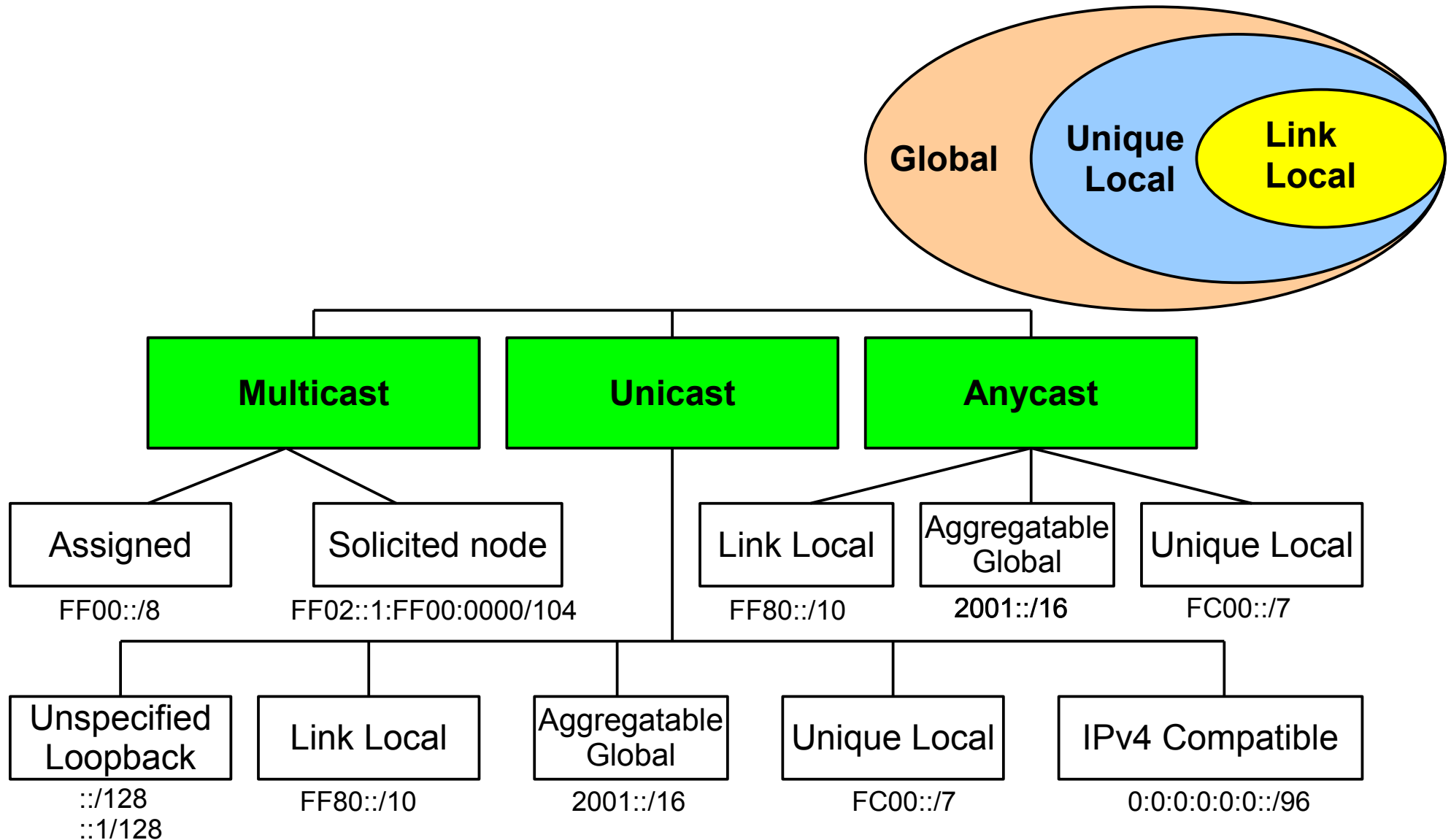


IPv6 Addressing Types

- Unicast
 - Identify one system on the Internet
 - Globally routable
 - Highest order bits are 001 (of Network Id)
- Multicast
 - Deliver to an entire group of systems
- Anycast
 - Deliver to any one of a group of systems
 - Ideal for mobile devices



Addressing Types





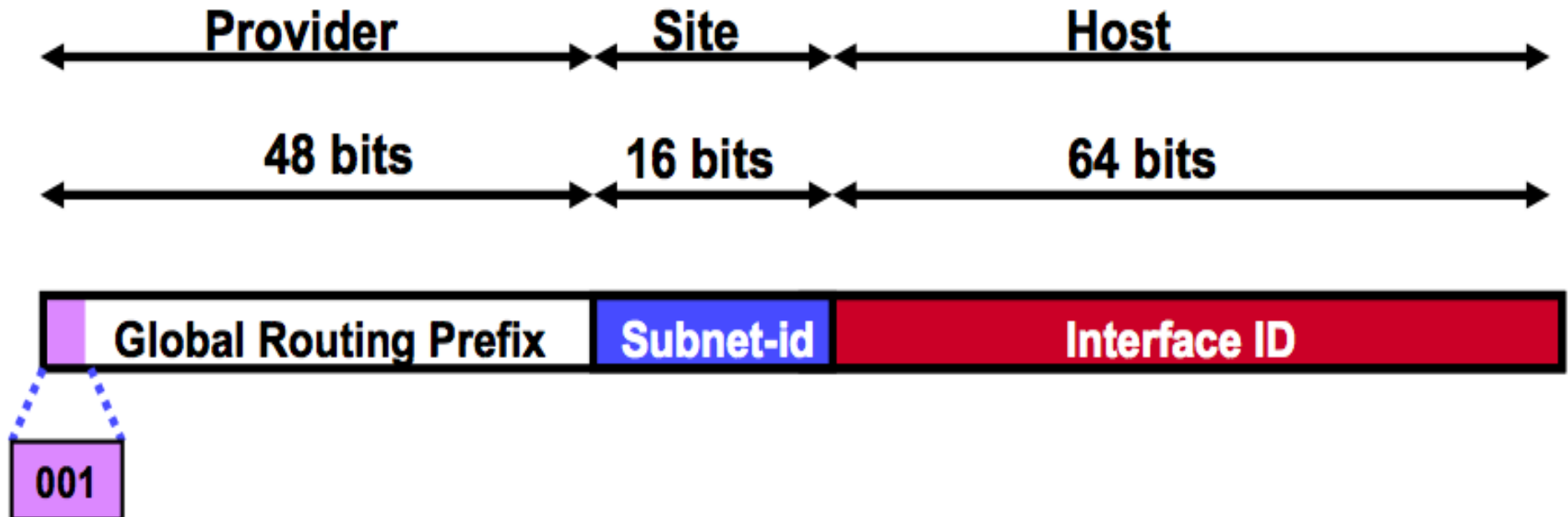
IPv6 Address Types (cont.)

Address Type	Binary Prefix	Prefix
unspecified	000...0 (128 bits)	::/128
loopback	000...01 (128 bits)	::1/128
link-local unicast	1111 1110 10	FE80::/10
multicast	1111 1111	FF00::/8
global unicast	All other addresses	



Unicast Addresses

- Global Unicast addresses are in 2000::/3 block
 - **200**1:5c0:1400:b::9773/128





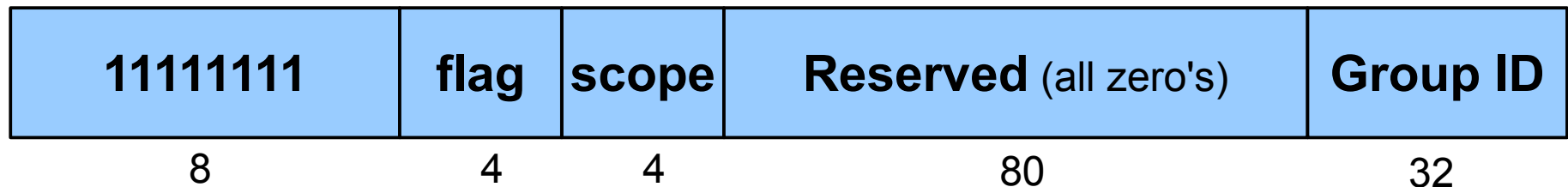
Anycast Addresses

- The same anycast address is assigned to a group of interfaces (nodes)
- However, a packet sent to an anycast address is delivered to the **nearest** one having this address
- Assigned from unicast address range
- Usage in the area of DNS discovery and Universal Plug and Play, but also used for multiple name, web and mail servers



Multicast Addresses

- In IPv6 multicast replaces IPv4 “broadcast”



- Identify a participating group of hosts
- Start with 0xFF (8 1-bits)
- One flag indicates transient (=1) or permanent (=0 or well-known address assigned)
- Must define a scope (global, site, link, node)
- Group ID: 1 = all nodes; 2 = all routers; etc



Multicast Scope

- A 4-bit field
- Likely values are
 - 1 : Node-local scope (interface)
 - 2 : Link-local scope (e.g. LAN)
 - 5 : Site-local (deprecated)
 - 8 : Organization-local scope
 - E : Global scope
- No broadcast address in IPv6, multicast to “*all nodes on the local link*” (scope 2; group-ID 1)

FF02::1



Well-known multicast group-numbers

Multicast Address	Meaning
FF02::1	All nodes on this link
FF02::2	All routers on this link
FF02::5	All OSPF routers on this link
FF02::9	All RIP routers on this link
FF02::1:2	All DHCP agents on this link
FF05::1:3	All DHCP servers on this link
FF05::101	All NTP servers on this link
FF02:0:0:0:1:FF::/104 combined with 24 low order bits from IPv6 address	Solicited-node multicast group (used to map MAC addresses)



Solicited node multicast addresses (for NDP)

- Multicast address built from unicast address
- Concatenation of FF02::1:FF00:0/104 and
 - 24 low order bits of unicast address (interface id)
- Nodes build their own IPv6 solicited node multicast address
- Nodes can use this technique to find of a destination host its MAC address, e.g.
 - 2001:001A:003F:1021:0100:0028:00**3F:0020**
 - **FF02:0000:0000:0000:0000:0001:FF**00:0000/104
 - **FF02:0000:0000:0000:0000:0001:FF3F:0020**
 - 33-33-**FF-3F-00-20** (multicast MAC address)



Neighbor Discovery Protocol

- Used to discover other hosts and routers on local network (stateless autoconfiguration)
- Makes use of the IPv6 multicast addresses (no ARP anymore)
- Uses ICMPv6 messages
 - Neighbor solicitation
 - Neighbor advertisement
 - Router solicitation
 - Router advertisement
 - redirect



Address Autoconfiguration Process

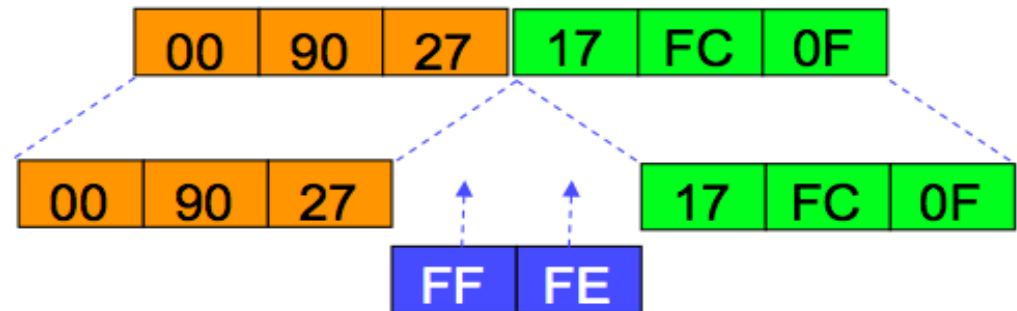
- Create a Link Local Address (FE80::/10)
 - No router or server required
- IPv6 address node configuration
 - Network ID
 - Manual
 - Auto (stateful or stateless)
 - Pre-defined well known prefix (link-local unicast FF80::/10)
 - Interface ID
 - Manual
 - Auto (stateful or stateless)



Link-Local Address

- Each interface has a Link-Local Address based on their MAC Address (IEEE EUI-64 - Extended Unique Identifier)

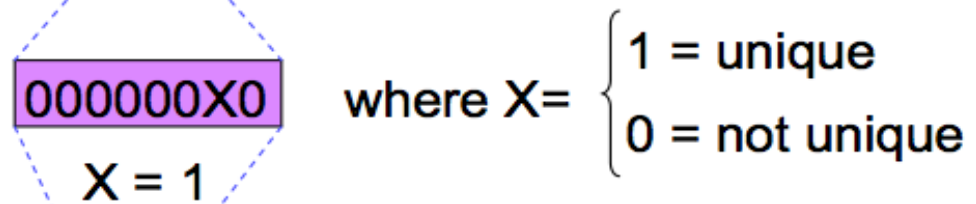
**Ethernet MAC address
(48 bits)**



64 bits version



Uniqueness of the MAC



Eui-64 address





Stateless Address Autoconfiguration

- Routers advertise prefixes that identify the subnet(s) associated with a link
- Hosts generate an "interface token" that uniquely identifies an interface on a subnet
 - Based on EUI-64 MAC address (security?)
 - Privacy Extensions:
`echo 1 > /proc/sys/net/ipv6/conf/all/use_tempaddr`
- An address is formed by combining the two



Router Solicitation (RS)

- Host sends a multicast Router solicitation when an interface is enabled
 - To discover IPv6 routers present on the link
 - To request an immediate Router advertisement
 - Sent to All-Router Multicast Address
 - Source link layer address of sender may be sent as an option
- IPv6 address
 - Source: unspecified (all zeros, `::/128`)
 - Destination: solicited-node multicast



Router Advertisement (RA)

- Router multicasts periodically (or on demand) its availability
- Router advertisements carry
 - Lifetime as a default router
 - Managed flag to inform hosts how to perform Address Autoconfiguration
 - List of prefixes used for a link
 - Link-layer address
 - Advertise an MTU for hosts to use on the link



Radvd daemon

- Stateless autoconfiguration with “router advertisement daemon (radvd)”

```
# cat /etc/radvd.conf
```

```
interface eth0
{
    AdvSendAdvert on;
    MinRtrAdvInterval 30;
    MaxRtrAdvInterval 100;
    prefix 2001:470:1f09:11b8::/64      # IPv6 address received for tunnel
    {
        AdvOnLink on;
        AdvAutonomous on;
        AdvRouterAddr off;
    };
};
```

```
# echo 1 > /proc/sys/net/ipv6/conf/all/forwarding
```



Stateful Address Autoconfiguration

- Clients obtain address and other optional parameters from DHCPv6 server
- DHCP server maintains the database and controls the address assignment
- Clients send DHCP solicit (DHCPv6 multicast address)
- Server responds with a DHCPv6 advertisement



DNS/Service Discovery @home

- How do I find my local file server?
- Multicast DNS (mDNS) = serverless DNS
 - DNS queries over IP Multicast in a small network where no DNS server is installed
 - Network prefix can change after modem reboots (no need to update /etc/hosts file!)
 - mDNS doesn't cross router boundary
- Service Discovery
 - DNS Service Discovery (mDNS/DNS-SD)
 - Universal Plug and Play (UPnP)



Multicast DNS (mDNS) @home



(1) mDNS Query to FF02::FB, port 5353,
Asking for AAAA record for **fileserverHome**



fileserverHome



(2) mDNS responder on
'fileserverHome' responds
To Multicast Group with
AAAA record



Implementations
Apple: Bonjour
Linux: Avahi



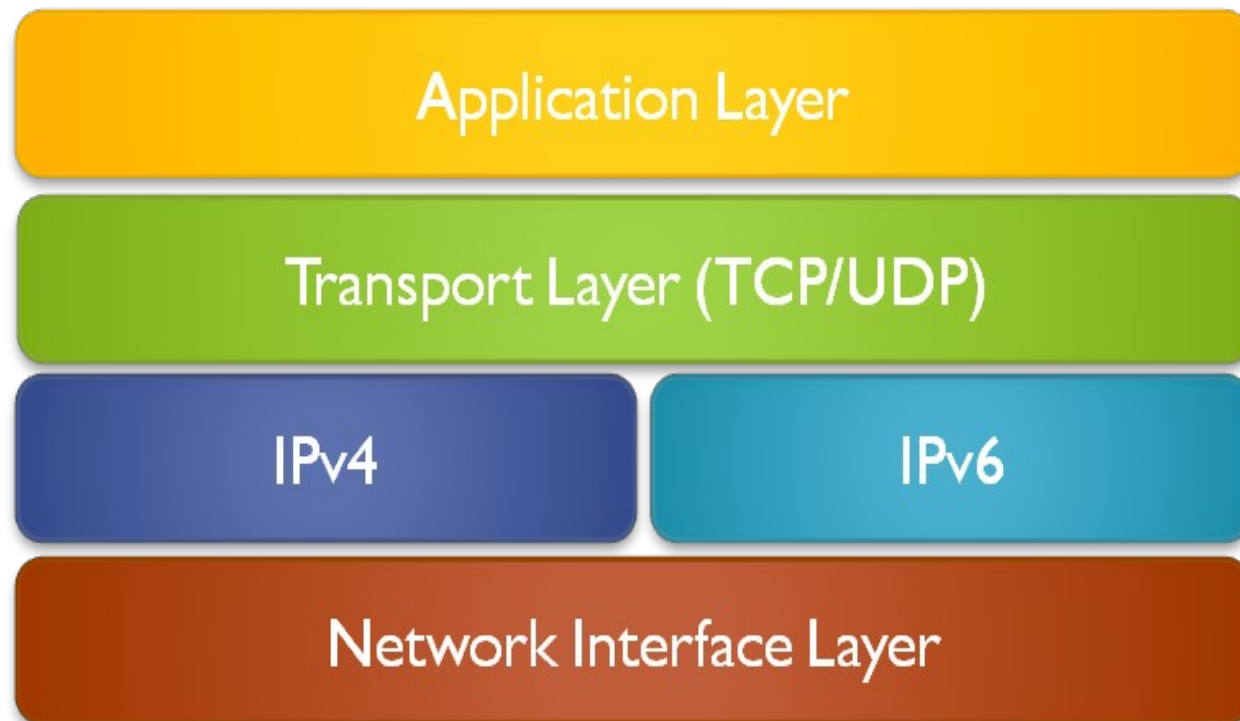
Transition Mechanisms

- Transition mechanisms are needed for IPv6 only host to reach IPv4 services.
- In the future we will see also IPv4 hosts need to be able to reach IPv6 services.
 - Dual Stack
 - Tunneling
 - Translation



Dual Stack

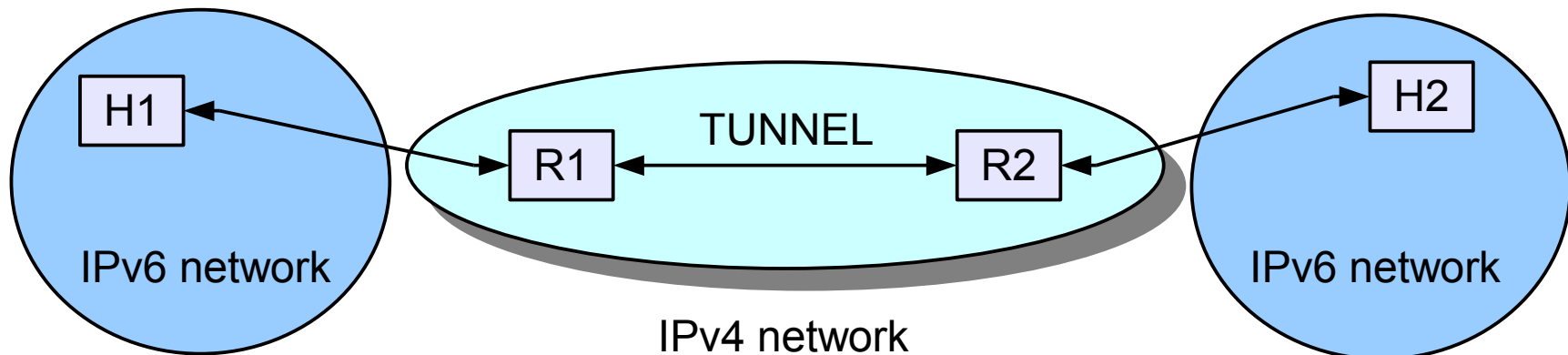
- Dual stack host can speak both IPv4 and IPv6
 - Communicate with IPv4 host by IPv4
 - Communicate with IPv6 host by IPv6



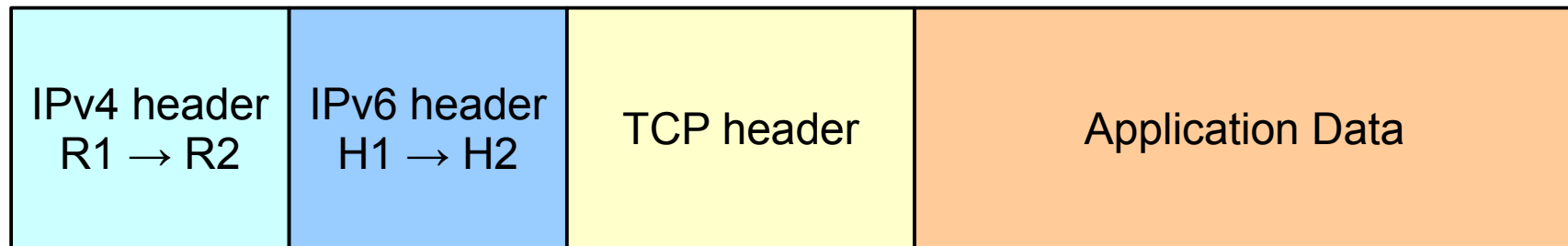


Tunneling

- Through an IPv4 tunnel we can connect two IPv6 networks
- Ideal to start experimenting with IPv6 topology



- Packet-structure with tunneling





Tunnel brokers

- There are 'free' tunnel brokers available
 - Require user registration
 - Request an IPv6 address (128 and 48 prefix)
 - Perfect to experiment with real IPv6 networking
- Hurricane Electric
 - <http://www.tunnelbroker.net/>
- SixXS
 - <http://www.sixxs.net/main/>
- GogoNET Freenet6
 - <http://gogonet.gogo6.com/>



Translation

- An extension to NAT techniques to translate header formats as well as addresses
- Translate IPv6 only host to IPv4 host (vice versa is not trivial)
 - Protocol translation
 - Mapping address
- Unreliable and try to avoid it



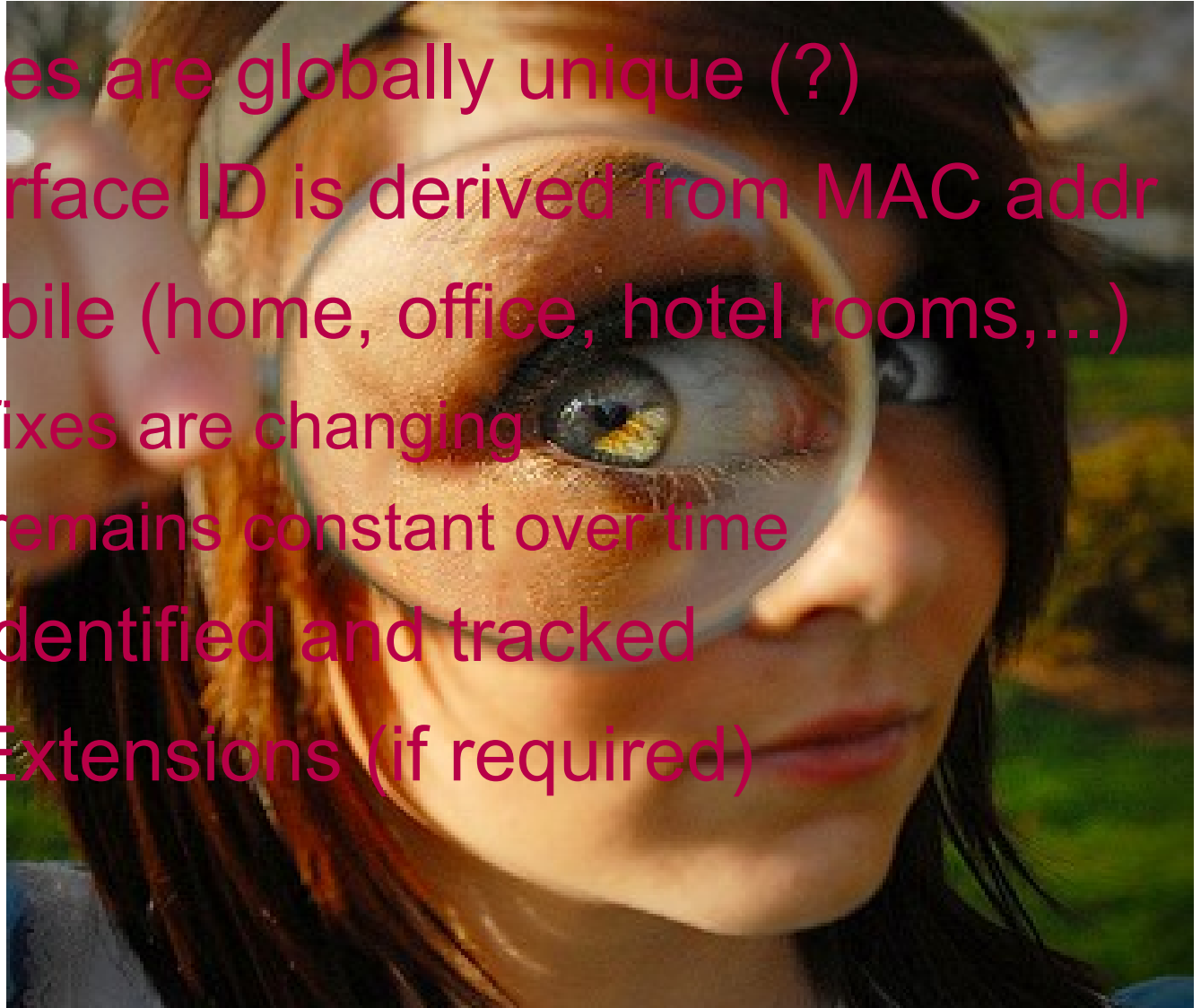
Security: protect yourself

- Once you start with IPv6 you must turn on ip6tables
- The radvd daemon will automatically configure interfaces on Windows (vista/windows7), Mac OS/X and Linux
 - Your IPv6 tunnel will open the gate to the IPv6 world
 - Attacker can send a Router Advertisement and gain access to your internal network (even you're safe on the IPv4 side)



Security Considerations

- MAC addresses are globally unique (?)
- SLAAC – Interface ID is derived from MAC addr
- Users are mobile (home, office, hotel rooms,...)
 - Network prefixes are changing
 - Interface ID remains constant over time
- User can be identified and tracked
- Use Privacy Extensions (if required)





How to become IPv6 ready?

- Buy only new equipment that is IPv6 compliant
- New software must be IPv6 capable
- Make an inventory of all current hard- and software
- Educate yourself via books, courses, and setup a lab environment
- Replace hard- and software where required
- Setup IPv6 DNS servers for public servers
- Get connected natively or via tunneling
- Use IPv6 for internal/external traffic (dual stack with IPv4)



Do and Don'ts

- Phased approach
- Change requirements for new hardware
- Work outside-in; then inside-out
- Dual stack; tunnels
- Think about possible future renumbering
- Don't separate IPv6 features from IPv4
- Don't do everything in one go
- Don't appoint an IPv6 specialist
- Don't buy from vendors unless they support IPv6



Make software IPv6 aware

- If you maintain an Open Source project invest time to make it IPv6 aware (if it uses IPv4 today)!
- Do what you preach:
 - Relax and recover (rear) is IPv6 ready since 1.11.0